

IT 220 – Business Data Communications

Fall 2024 Final Course Project

Multilocation Restaurant Group Corp

Christos Paloubis

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Organizational Overview and Project Description

This network design project is for a fictitious organization named Multilocation Restaurant Group Corp., a company operating in the hospitality industry with two restaurant locations, a home office, and a corporate office. Each restaurant location features two VLANs: one dedicated to the POS (Point-of-Sale) network for secure transaction processing and another for the computers and WiFi system used by staff and guests. The home office is connected to all networks for centralized management and operations, while the corporate office serves as the administrative hub, linking all locations into a cohesive network.

The purpose of this project is to design a functional and scalable network for Multilocation Restaurant Group Corp. using Cisco Packet Tracer. The network consists of four individual LANs the 14th Street restaurant, the Columbus Ave restaurant, the home office, and the corporate office. They are interconnected with dynamic routing using OSPF to ensure seamless communication. Each LAN is tailored to the specific needs of its location. The network uses IP addressing across Class A, B, and C networks, ensuring scalability and efficient resource utilization. The design prioritizes security, reliability, and operational efficiency to meet the needs of the organization.

14th Street Location

- 8 POS Terminals
- 1 POS Server
- 2 Computers
- 11 Remote IP Printers
- 3 Wireless Tablets
- 2 Wireless Access Points

Home Office

- 2 Computers
- 1 Laptop

Corporate Office

- 3 POS Terminals
- 2 Tablets
- 1 Smart Phone

Columbus Ave

- 4 POS Terminals
- 1 POS Server
- 2 Computers
- 4 Remote IP Printers
- 1 Wireless Access Points
- 1 Wireless Tablet